SET 3 – HOSPITAL APPOINTMENT BOOKING APPLICATION

Agile and Scrum Assignment Report | N. Harshith | Jira Project: HAB

Business Scenario

A hospital group wants a mobile application for booking, rescheduling and cancelling doctor appointments, with reminder notifications. The application also supports doctor schedule management.

Task 1 – Product Vision and Stakeholder Mapping

Product Vision: The Hospital Appointment Booking Application provides patients with a convenient way to find doctors, view available appointment slots, book appointments, reschedule or cancel bookings, and receive timely reminders. It also helps doctors manage schedules and view daily appointments, reducing manual coordination and improving the patient experience.

Stakeholder mapping

Stakeholder	Scrum role mapping	Main responsibility
Patients	Customer	Search doctors, book, reschedule and cancel appointments.
Hospital management / product representative	Product Owner	Sets product vision, prioritises backlog and accepts completed work.
Scrum facilitator	Scrum Master	Facilitates Scrum events and removes team impediments.
Developers (frontend, backend, QA)	Developers	Design, build, test and integrate application features.
Doctors	Customer / stakeholder	Provide schedule requirements and use appointment lists.
Reception / appointment staff	Customer / stakeholder	Provide booking workflow feedback and operational needs.
IT / security / compliance staff	Stakeholder consulted	Advise on access, privacy, security and system constraints.

The Product Owner, Scrum Master and Developers are the core Scrum accountabilities. Patients, doctors and hospital staff provide customer and stakeholder input.

Task 2 – Epics and User Stories

Four Epics and twelve User Stories were created in Jira under the Hospital Appointment Booking project (HAB).

Epic	Jira ID	User Story
Appointment Booking	HAB-5	Search for available doctors.
Appointment Booking	HAB-6	View doctor profiles and specializations.
Appointment Booking	HAB-7	View available appointment slots.
Appointment Booking	HAB-8	Book an appointment.
Rescheduling and Cancellation	HAB-9	View booked appointments.
Rescheduling and Cancellation	HAB-10	Reschedule an appointment.
Rescheduling and Cancellation	HAB-11	Cancel an appointment.
Reminders and Notifications	HAB-12	Receive booking confirmation.
Reminders and Notifications	HAB-13	Receive appointment reminders.
Reminders and Notifications	HAB-14	Receive cancellation or rescheduling notifications.
Doctor Schedule Management	HAB-15	Doctor views daily appointments.
Doctor Schedule Management	HAB-16	Doctor manages available time slots.

All 12 stories are assigned to their corresponding parent Epics in the Jira backlog.

Task 3 – INVEST Review

Element	Review applied
Independent	Stories represent distinct capabilities. The booking flow uses separate stories for doctor search, profiles, slots and booking.
Negotiable	Implementation details can be discussed with the Product Owner and team.
Valuable	Each story provides value to patients, doctors or hospital operations.
Estimable	Stories have clear outcomes and relative story-point estimates.
Small	Stories focus on one primary action; larger capabilities can be split further if needed.
Testable	Each story can be verified through observable application behaviour and acceptance criteria.

Refinement: Separating doctor search, profile viewing, slot viewing and booking keeps the booking journey understandable and makes each capability easier to estimate and test.

Task 4 – Given / When / Then Acceptance Criteria

HAB-5 – Search for available doctors
Given: The patient is logged in.
When: The patient searches by doctor name or specialization.
Then: The system displays matching doctors.

HAB-7 – View available appointment slots
Given: The patient has selected a doctor.
When: The patient views the doctor's schedule.
Then: The system displays available appointment slots.

HAB-8 – Book an appointment
Given: The patient selects an available slot.
When: The patient confirms the booking.
Then: The system books the appointment and displays confirmation.
And: If the slot has already been taken, the system displays an error and asks the patient to select another slot.

HAB-11 – Cancel an appointment
Given: The patient has a confirmed appointment.
When: The patient cancels the appointment.
Then: The system cancels the appointment and updates its status.

Task 5 – Prioritisation and Planning Poker

The backlog was prioritised according to business value and relative effort. High-priority work supports the core patient booking journey and doctor scheduling; supporting information and notification features are medium priority.

Jira ID	Story	Priority	Points
HAB-5	Search for available doctors	High	3
HAB-6	View doctor profiles and specializations	Medium	2
HAB-7	View available appointment slots	High	3
HAB-8	Book an appointment	Highest	5
HAB-9	View booked appointments	Medium	2
HAB-10	Reschedule an appointment	High	3
HAB-11	Cancel an appointment	Medium	2
HAB-12	Receive booking confirmation	High	2
HAB-13	Receive appointment reminders	Medium	3
HAB-14	Receive cancellation/rescheduling notifications	Medium	2
HAB-15	Doctor views daily appointments	High	3
HAB-16	Doctor manages available time slots	High	5
	TOTAL		35

Planning Poker: Team members discuss each story, clarify scope, reveal Fibonacci estimates together, discuss differences and agree on a final relative estimate. The backlog total is **35 story points**.

Story points versus hours: Hours estimate elapsed time. Story points estimate relative effort, complexity, uncertainty and risk. A 5-point story is more complex or uncertain than a 2-point story; it does not mean five hours.

Task 6 – Jira Backlog, Definition of Ready and Definition of Done

The 12-story backlog was published in the HAB Jira project. The team checklists were created and published in Confluence under “Hospital Appointment Booking – DoR and DoD.”

Definition of Ready (DoR)

- User story has a clear description.
- Acceptance criteria are defined.
- Story follows INVEST principles.
- Dependencies are identified.
- Priority is assigned.
- Story points are estimated.

Definition of Done (DoD)

- Development is completed.
- Code is reviewed and approved.
- Unit tests pass successfully.
- All acceptance criteria are satisfied.
- No critical bugs remain.
- Documentation is updated.

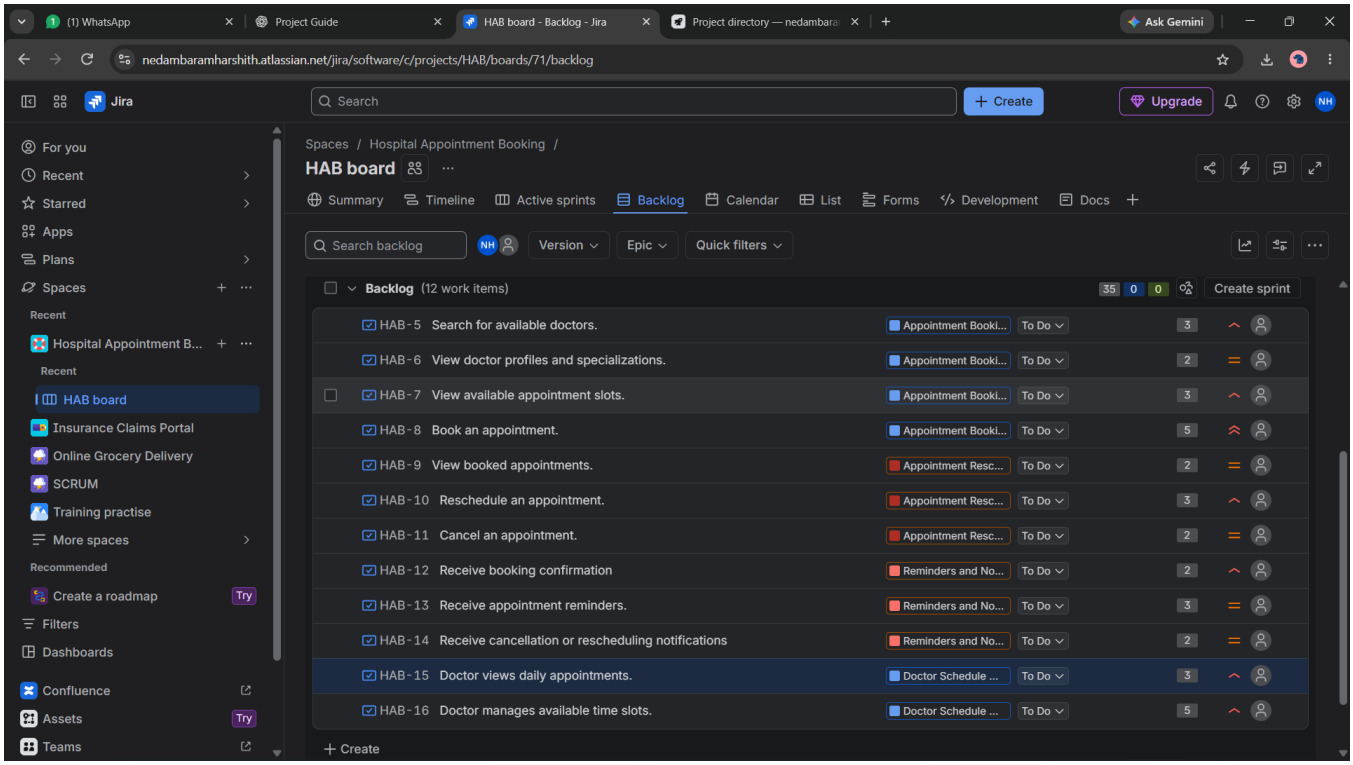
Purpose: DoR ensures stories are prepared before development. DoD ensures completed work meets agreed quality standards and is ready for delivery.

Jira and Confluence Evidence

Actual screenshots supplied during the implementation are included below.

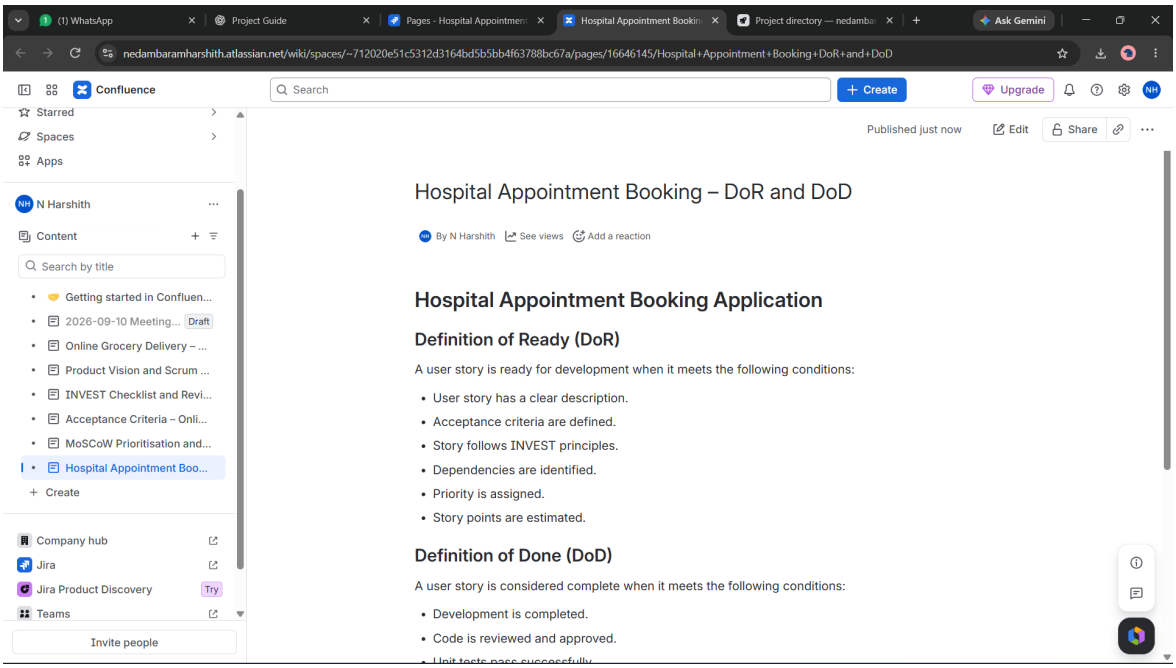
Jira backlog – 12 User Stories and 35 Story Points

Shows the 12 stories, Epic assignments, issue IDs, priority indicators and estimates.



Confluence – Definition of Ready

Shows the published DoR and DoD page and the six DoR checklist items.



Confluence – Definition of Done

Shows the DoD checklist and purpose statement.

